

MARY AMOAH

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PROFILE

Cybersecurity professional with experience across security operations, incident response, threat hunting, detection engineering, security automation, vulnerability management, and cybersecurity research. Experienced in designing and optimizing SIEM platforms, investigating security incidents, integrating threat intelligence, and developing automated workflows that improve detection accuracy and response efficiency. Skilled in enterprise security technologies, security analytics, security engineering, and AI-assisted cybersecurity solutions. Passionate about advancing cyber defense capabilities through automation, continuous improvement, and applied research.

CORE COMPETENCIES

Security Operations • Incident Response • Threat Hunting • Detection Engineering • Security Automation • Threat Intelligence • Vulnerability Management • Security Research • AI & Machine Learning

KEY ACHIEVEMENTS

- Led deployment, administration, and optimization of a Wazuh SIEM platform supporting 180+ enterprise log sources across network, endpoint, server, and application environments.
- Reduced false positives by approximately 85% through detection engineering, parser optimization, log normalization, and alert correlation improvements.
- Designed and implemented automated investigation, enrichment, and response workflows using Python, n8n, and security platform integrations.
- Deployed and integrated OpenCTI, TheHive, Cortex, and Wazuh into a centralized threat intelligence and incident response ecosystem.
- Developed AI-assisted SOC investigation workflows that transform raw security alerts into structured analyst-ready investigations and recommendations.
- Designed and developed a Security Operations Reporting and Management Platform to improve SOC visibility, reporting, operational metrics, SLA tracking, and management decision-making.
- Built custom integrations with security technologies including Palo Alto, FortiGate, F5 WAF, VirusTotal, AbuseIPDB, OpenCTI, Cortex, and TheHive.
- Conducted security assessments, vulnerability validation, and security reviews for enterprise applications and infrastructure before production deployment.

PROFESSIONAL EXPERIENCE

Cybersecurity Specialist

Sultan Qaboos University, Muscat, Oman | 2025 – Present

- Lead security monitoring, incident investigation, and response activities across a large enterprise environment supporting critical academic and administrative services.
- Administer and optimize a Wazuh SIEM deployment monitoring more than **180 log sources** across servers, endpoints, firewalls, web applications, and infrastructure systems.
- Develop and tune detection rules aligned with MITRE ATT&CK, improving detection coverage and reducing false positives by approximately 85%.
- Conduct proactive threat hunting and complex investigations involving network, endpoint, authentication, and web application security events.
- Design and implement security automation workflows using Python and n8n for alert enrichment, threat intelligence correlation, case creation, and response actions.
- Deploy and integrate Wazuh, TheHive, Cortex, and OpenCTI into a centralized security operations and threat intelligence ecosystem.
- Design and develop a Security Operations Reporting & Management Platform to improve operational visibility, reporting, SLA tracking, and SOC performance measurement.
- Build AI-assisted investigation workflows that transform security alerts into structured investigation summaries and analyst recommendations.
- Conduct vulnerability validation, security assessments, and security reviews before production deployment of enterprise applications.
- Serve as an escalation point for high-severity incidents and provide mentorship to junior analysts and interns.

Cybersecurity Analyst

Virtual Infosec Africa, Ghana | 2023 – 2025

- Performed security monitoring, alert triage, and incident investigations using SIEM technologies.
- Conducted root cause analysis for security incidents across network, endpoint, and identity platforms.
- Assisted with vulnerability assessments and remediation validation activities.
- Supported threat hunting and detection engineering initiatives.

SOC Analyst

Cibera Defence, India | 2022 – 2024

- Assisted with log analysis, threat detection, and security monitoring operations.
- Investigated authentication, network, endpoint, and application security events.
- Supported incident response and escalation activities.

Cybersecurity Intern

TechDefence Labs, India | 2022

- Assisted with penetration testing and vulnerability assessment activities.
- Supported cybersecurity research and security simulation exercises.

TECHNICAL SKILLS

- **Security Operations:** Incident Response, Threat Hunting, Detection Engineering, Security Monitoring, Threat Intelligence
- **SIEM & Security Platforms:** Wazuh, Microsoft Sentinel, Splunk, LogRhythm, FortiSIEM, TheHive, Cortex, OpenCTI
- **Security Automation:** Python, Bash, SQL, Java, n8n, API Integrations
- **Network & Infrastructure Security:** Palo Alto, FortiGate, F5 WAF, Trend Micro, Linux, Docker, Git
- **AI & Security Analytics:** Machine Learning, Anomaly Detection, Security Analytics, Pandas, scikit-learn
- **Security Practices:** Vulnerability Assessment, Security Validation, MITRE ATT&CK, NIST CSF, CIS Controls

SELECTED SECURITY PROJECTS

Security Operations Reporting & Management Platform

Developed a centralized platform for SOC reporting, SLA tracking, operational metrics, and security management.

AI-Assisted SOC Investigation Framework

Built AI-driven workflows to automate alert analysis and generate structured investigation summaries.

SOC Automation & Response Platform

Implemented automated enrichment, correlation, and response workflows using Wazuh, Cortex, Python, and n8n.

Threat Intelligence Platform Integration

Deployed and integrated OpenCTI, TheHive, Cortex, and Wazuh to support intelligence-driven security operations.

Security Telemetry Anomaly Detection

Developed machine learning models to identify anomalous behavior within security telemetry datasets.

EDUCATION

MSc Cybersecurity and Forensics

Gujarat University, India | **Distinction**

BSc Forensic Science

University of Cape Coast, Ghana | **First Class**

CERTIFICATIONS

- EC-Council Certified SOC Analyst (CSA)
- EC-Council Certified Incident Handler (ECIH)
- LogRhythm Security Analyst
- LogRhythm Platform Administrator

LANGUAGE SKILLS

English: Native proficiency

French: Conversational proficiency